TED(21)-5073B **REVISION 2021**

MODEL QUESTION PAPER

MODERN SEPARATION TECHNIQUES

Max. Marks : 75

Part A

I. Answer all questions in one word or one sentence. Each question carries 1 mark

9 X 1 = 9 Marks

1. Write the principle of liquid chromatography.
2. Define chemisorption.
3. Name any two areas in which membrane separation process is used.
4. Name any two methods of membrane manufacture.
5. What is the driving force in pervaporation.
6. What are isotonic solutions.
7. Name the compounds used for softening hard water.
8. List two applications of cryogenic fractionation.
9. List any two types of liquid membranes.

Part B

II. Answer any EIGHT questions from the following. Each questions carries 3 marks

8 X 3 = 24 Marks

1. Explain the principle of Isopycnic centrifugation.
2. Write a brief note on gradient chromatography.
3. What are the methods adopted in cleaning the membranes.
4. List the applications of plate and frame membrane module.
5. Briefly explain the track etch method of membrane manufacture.
6. Sketch the configuration of a typical ultrafiltration unit.
7. List the factors to be considered for the selection of RO membranes.
8. Differentiate permeation and pervaporation.
9. Enumerate the benefits of liquid membranes.
10. Write short note on micellar separation.

Part C

Answer ALL questions. Each questions carries 7 marks

6 X 7 = 42 Marks

III. With a clear sketch explain the working of a rotary bed adsorber.

OR

IV. Explain the gas chromatographic technique and give applications.

V. Explain about the various types of synthetic membranes.

OR

VI. List the differences between hollow fibre and spiral wound membrane modules.

VII. With a neat sketch explain the sol gel peptization method.

OR

VIII. Explain the manufacture of ion exchange membranes. Give examples.

IX. Explain in detail the osmotic pinch effect.

OR

X. Briefly explain the types of devices used in ultra filtration.

XI. With a neat sketch explain electrodialysis.

OR

XII. Explain the cross flow and counter current models of gas separation using membranes.

XIII. List the characteristics of polymer inclusion membranes. What are its advantages.

OR

XIV. Differentiate the froth flotation and foam fractionation methods.